

CLIENT: DISPOSABLE

VERSION: 0.1

DATE: 2026-07-15

OWNER: ENGINEERING

A structural alignment agreement for Foley Balloon Catheter line quality tracking.

🕒 How to Use This Document

Everything marked **[A#]** is an operational assumption derived from your current Excel books. Please mark each assumption: ✓ **Confirm**, ✗ **Wrong (write the correct rule)**, or ? **Unsure**. Once complete, click **Generate Response** at the bottom to export the text block for email or system sign-off.

01. What is one “fact” we store?

Every number in the system is stored as a single, granular **observation** to prevent estimates or double-counting:

FIELD	MEANING	EXAMPLE FROM YOUR WORLD
When	Business date the inspection/production happened	2025-04-02
Stage	Process / inspection gate	Visual Inspection, Balloon Testing, Valve Integrity, Final
Size	French size (when sheet is size-wise)	16 FR
Batch	Work unit / lot identifier (when present)	25A28 (seen on VISUAL-style sheets)
Measure	What was counted	Checked / Accepted / Hold-Rework / Rejected
Defect	Rejection reason (when broken down)	COAG, PINH, STBL, LEAK, OTH, ...
Qty	Integer count (pcs; sometimes trolleys)	120
Source	Where the number came from	File name + sheet + cell, or manual entry

System Rule: We do not invent or interpolate numbers. Missing data fields show as blank/empty, never estimated or filled by default.

02. Dimensions & Batch Definition

Assumptions establishing what "batch" and process tracking variables represent:

ID	ASSUMPTION DETAILS	CONFIRM / CORRECT
[A1]	A batch is a lot / batch number written on the inspection sheet (e.g. <code>25A28</code>), not a trolley and not "the whole day." Write definition if wrong...	✓✗?
[A2]	One batch can appear on more than one stage (e.g. same lot later at Balloon / Final) — stages are separate facts, not one merged row. Write rule if wrong...	✓✗?
[A3]	Batch is optional. Many rows in your books are day-only (no batch column). Those stay valid with <code>batch = (none)</code>. Write rule if wrong...	✓✗?
[A4]	Size and batch are independent: same day can have multiple sizes and multiple batches. Write rule if wrong...	✓✗?
[A5]	Shift (Day/Night) is metadata for entry/audit, not the primary KPI axis (unless you say otherwise). Write rule if wrong...	✓✗?

ID	ASSUMPTION DETAILS	CONFIRM / CORRECT
	Machine / operator / supervisor are optional tags, not required for rejection rate math.	✓ ✗ ?
[A6]	Write rule if wrong...	

If [A1] is wrong, write your definition here:

Batch means...

03. Aggregation & Math Rules

Decisions that establish how metrics roll up and how data discrepancies are handled:

ID	SITUATION DETAILS & DEFAULT RULE	CONFIRM / CORRECT
[A7]	Same stage + day + size, different batches: Keep as separate batch rows. For daily/monthly KPIs, sum them. Write rule if wrong...	☒ ☐ ☐
[A8]	Same stage + day, one file has per-size rows and also a whole-line total: Prefer size rows (sum = stage total); drop the redundant whole-line total to avoid double count. Write rule if wrong...	☒ ☐ ☐
[A9]	Same stage + day described by two different uploaded files: Keep the higher-precedence source only (size-wise / detailed > summary). Direct manual entry wins over upload for that day. not always the direct manual entry wins but depends on how detailed the entered data is when compared to the uploaded excel	☐ ☒ ☐
[A10]	Re-upload of the same file: **Idempotent** — does not double-count. Write rule if wrong...	☒ ☐ ☐
[A11]	Sheet has a % rejection formula cell: Store as a claim to verify, not as the source of truth. Dashboard recomputes rate from checked & rejected.	☒ ☐ ☐

ID	SITUATION DETAILS & DEFAULT RULE	CONFIRM / CORRECT
	Write rule if wrong...	
[A12]	Defect columns sum $\neq$ Rejected total: Flag as a data issue (warning); still store both; do not silently “fix” the sheet. flag it as data issue (warning) and then attempt to fix tht issue according to the calulated value with formula logically and ask for approval to change it	✓ X ?
[A13]	When a manager opens “Batch view,” should they see unmerged batch lines (even if same day/size)? (Default: YES) Provide details if not default...	✓ Yes X No ?
[A14]	When a manager opens “Monthly dashboard,” should they see sums across batches for that month? (Default: YES) Provide details if not default...	✓ Yes X No ?

04. Primary Views & Dashboards

Views designed for first deployment priority (v1 optimization):

PRIORITY	VIEW	QUESTION IT ANSWERS	DEFAULT STATE
P0	Stage · Day	"What happened at Visual yesterday?"	ON
P0	Stage · Day · Size	"Which FR is rejecting more this week?"	ON (where size-wise sheets exist)
P0	Defect Pareto (scoped by stage/period)	"Which defect codes dominate?"	ON
P0	Batch table	"How did lot 25A28 perform?" / "Worst batches this week?"	ON — elevated
P1	Week / Month / FY rollups	Management calendar reporting	ON
P2	SPC control charts	Process stability	After P0 trusted
P2	COPQ ₹ savings	Cost of poor quality	After P0 trusted
P3	Full 27-step process map	Line storytelling	Optional overlay

[A15] The Monday-morning default home screen should be:

✓ Agree

X Modify

?

Factory overview

Displays all quality gates for the current period (Recommended default)

Batch list

Show latest production batches first for lot review

Single stage deep-dive

Focus closely on a specific inspection point

Other layout



Specify other layout requirements...

05. Process Stages & French Sizes

The gates and sizing parameters defined for mapping:

QUALITY GATE (STAGE)	SIZE-WISE?	DEFECT BREAKDOWN?	TYPICAL SOURCE DOCUMENT / BOOK
Visual Inspection	Yes	Yes (COAG, PINH, ...)	VISUAL / size-wise books
Balloon Testing	Yes	Yes (STBL, BLBR, LEAK, ...)	Valve / balloon books
Valve Integrity	Yes	Yes (LEAK, 90/10, BUB, THSP, ...)	Same family books
Final Inspection	Yes	Yes (subset of visual-like codes)	FINAL sheets
Production / eye-punching / ...	Often no	Often no or light	Daily activity / shopfloor log

[A16] These four gates (Visual, Balloon, Valve Integrity, Final) are the **primary** analytics universe for v1; other process steps are secondary until you prioritize them.

✓
Confirm

✗
Wrong

?

Write rule or list other gates...

[A17] French sizes **6–24 FR** cover production. Confirm the active sizes or deselect unused ones below:

✓
Confirm

✗
Wrong

?

Provide sizing notes...

6 FR

8 FR

10 FR

12 FR

14 FR

16 FR

18 FR

20 FR

22 FR

24 FR

[A18] Fiscal year starts **April** (FY Apr–Mar).

✓ Confirm

X Wrong

?

Write different FY period...

06. Source of Truth (Conflict Resolution)

Rules that determine what wins when data entries disagree:

ID	CONFLICT SITUATION & RESOLUTION RULE	CONFIRM / CORRECT
[A19]	Excel upload vs manual data entry for same stage-day: Manual entry wins.	✓ X ?
	depends on how much detailed the data is available if the manual data is more detailed and specific prefer it else prefer the excel	
[A20]	Detailed size-wise book vs monthly summary book: Detailed book wins.	✓ X ?
	Write rule if wrong...	
[A21]	Two "official" numbers from different departments: System flags conflict ; human adjudicates — we do not pick silently.	✓ X ?
	Write rule if wrong...	
[A22]	Column header spelling variants ("STRUCK BALLOON" / "BALLOOM BRUST"): Map to one defect code after you verify mappings once; keep raw text for audit.	✓ X ?
	Write rule if wrong...	

07. MLP Scope & Priority Ranking

Features slated for the initial Minimum Viable Product (MLP) launch versus later updates:

✔ In Scope (v1 Standard)

- 1. Workbook uploads with column mapping (numbers match sheet perfectly).
- 2. Day and Batch views (where batch numbers are captured).
- 3. Stage, size, and defect breakdowns with dynamic "View Source" provenance.
- 4. Zero silent double counting (idempotent uploads).
- 5. Export & audit system showing raw entries.

🕒 Explicitly Later (Post-v1)

- Full SPC suite statistical polish.
- COPQ finance-grade cost structures.
- CAPA ticket automation.
- Multi-plant / multi-product data mapping.
- AI natural language chat interface.

[A23] Do you agree with this MLP feature order?

✔ Agree

✕ Modify

?

If you would like to prioritize differently, rank your top 3 concerns below:

RANK 1 PRIORITY

e.g. Workbook upload integrity

RANK 2 PRIORITY

e.g. Batch tracking

RANK 3 PRIORITY

e.g. Excel audits

08. Worked Example & Layout Preference

A concrete example mapping raw inputs to expected system views:

Source Sheet Rows (Visual Stage, 2025-04-02)

BATCH	SIZE	CHECKED QTY	REJECTED QTY	PINH DEFECT QTY
25A28	16 FR	1,000	12	5
25A29	16 FR	800	4	1
25A28	18 FR	500	6	2

Default Display Resolution Mapping

TARGET VIEW	EXPECTED AGGREGATION RESULT
Batch 25A28 View	Two lines (16 FR + 18 FR): checked 1500 (1000+500), rejected 18 (12+6)
Day 2025-04-02 · Visual · 16 FR	Checked 1,800, Rejected 16 (both batches summed together)
Monthly Rollup	Sums all days (batches are rolled up unless Batch view is selected)

[A24] For one batch with multiple sizes, default is keep size split inside the batch.

Do you prefer instead one total per batch (hiding size details)?

Write layout request...

✓ Keep Size Split

X Rollup / Hide Size

?

09. Client Sign-off Panel

Confirm authorization and sign-off on the operational logic detailed above:

Instructions for Client: Enter your name, date, and confirm your approval below. Assumptions marked with **X Wrong** will be compiled into the final PRD rules list for modification.

Plant / Quality Owner

FULL NAME

Authorized Signatory Name

DATE

dd-mm-yyyy

CONFIRM / SIGN (TYPE OK)

Type OK to confirm

Daily Dashboard User (Quality Lead)

FULL NAME

User Name

DATE

dd-mm-yyyy

CONFIRM / SIGN (TYPE OK)

Type OK to confirm

Excel Book Owner / Data Entry Lead

FULL NAME

Data Owner Name

DATE

dd-mm-yyyy

CONFIRM / SIGN (TYPE OK)

Type OK to confirm

Engineering Representative

FULL NAME

Lead Engineer Name

DATE

dd-mm-yyyy

CONFIRM / SIGN (TYPE OK)

Type OK to confirm